

Optimizing Sales Forecasting and Inventory Management for Mr. Pizza shop

Mid Term report for the BDM capstone Project

Submitted by

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1. Executive Summary

Mr. Pizza is a small, independently owned pizza outlet located in Kathgodam, Uttarakhand, established in 2023 and managed solely by Mr. Prakash Singh Danu. The shop operates as a B2C food retail business offering pizzas, burgers, hot dogs, breads, shakes, desserts, and beverages through walk-in customers and the Zomato online platform. Occasional event catering services are also provided.

The primary business problems identified are: (1) inability to accurately forecast daily sales demand, leading to ingredient stockouts and wastage; (2) absence of systematic inventory management, resulting in over-purchasing of low-demand items; and (3) lack of insight into customer order trends for retention planning. These problems were identified through direct interaction with the owner during data collection visits.

To address these, primary data was collected directly from the owner's handwritten sales notebook covering five months (September 2023 to January 2024), comprising 997 line-item records across 139 trading days. The dataset captures item-level sales including item name, category, size, quantity, unit price, and daily revenue. Key findings include: Pizza accounts for 79.7% of total revenue (₹1,52,977); Sunday and Saturday are the highest-revenue days contributing over 41.7% of weekly revenue; January 2024 shows the strongest revenue recovery after a winter dip in December 2023; and Mix Veg Margharita Pizza and Kulhad Pizza are the top two revenue-generating items. These findings will be used in the final submission for demand forecasting and ABC inventory classification.

2. Proof of Originality

The data for this project has been collected through direct, in-person visits to Mr. Pizza, Kathgodam, Uttarakhand. The following evidence of originality is provided:

- **Video Interaction:** A video recording of an interaction with the owner, Mr. Prakash Singh Danu, is available. The video covers an introduction to the business, the problems faced in daily operations, and his expectations from this project.
<https://youtu.be/O2RF-AWu43k>
- **Photographs:** Images and videos of the shop, owner, menu board, and the handwritten sales notebook are attached here.
https://drive.google.com/drive/folders/1X6-ZVyTAp8wnHbVAjNr76p6O8RJztaYw?usp=drive_link
- **Letter from Organization:** A signed letter from Mr. Prakash Singh Danu on behalf of Mr. Pizza, confirming my engagement with the business for this project.
- **Google Drive Link (Photos, Video, Letter):** **BDM Project**

3. Data Collection Process

3.1 Field Notes

The data collection process began with an initial meeting with the owner of Mr. Pizza to understand the business operations, identify recurring problems, and assess available data sources. The owner indicated that the primary record-keeping method was a handwritten notebook in which daily sales were recorded by item name, size, quantity, and daily total.

Subsequent visits were made to digitize this notebook data over multiple sessions.

The key business problems identified during these interactions were: (1) inability to predict daily demand leading to ingredient stockouts, particularly for dairy products such as cheese and milk; (2) lack of insight into which menu items drive the most revenue, making it difficult to prioritize inventory purchasing; and (3) no systematic tracking of customer order patterns for retention purposes.

3.2 Data Collection Method

Primary data was collected directly from the owner's handwritten sales notebook. The notebook records the following information for each transaction: date, item name, size of the item, quantity ordered, and the final daily total. This data was manually transcribed into a structured Google Sheets template designed specifically for this project, with VLOOKUP formulas to auto-fill unit prices from a Lookup sheet. The data spans five months from September 2023 to January 2024, covering 139 trading days and 997 individual line-item transactions.

3.3 Data Cleaning and Pre-processing

- **Item name standardization:** Inconsistent spellings were standardized (e.g., 'Thandoori' vs 'Tandoori'; 'Peppy Paneer Pizza' vs 'Peppy Paneer').
- **New/unlisted items:** Items not present on the printed menu (Extra Cheese, Choco Lava Cake, Cold Drink, Water, Bun, Chicken Pizza) were retained in the dataset and assigned appropriate categories as they represent genuine sales.
- **Cross-verification:** Sum of line totals for each date was compared against the owner's handwritten daily total. Zero discrepancies were found across all 30 days where a daily total was recorded.
- **Acknowledged limitations:** Purchase data is only available for cheese, milk, and toppings (no wheat/flour records). Cash vs. UPI payment data was available for only 2–3 days and was excluded. Missing dates in the notebook were left as gaps and not filled with zero values.

4. Metadata

Metadata refers to data that describes the structure, content, and context of the collected dataset. The table below documents each variable in the sales dataset, its data type, a description of what it captures, and its direct relevance to the problem statement. The dataset contains 11 variables across 997 observations.

Table 1: Metadata of the Mr. Pizza Sales Dataset

Variable Name	Data Type	Description	Relevance to Problem	Sample Values
Date	Date (DD-MM-YYYY)	Calendar date of each sales transaction	Enables time-series and seasonal trend analysis	01-09-2023

Day	Categorical (Text)	Day of week derived from Date	Identifies peak days for inventory and staffing planning	Monday, Sunday
Category	Categorical (Text)	Broad product classification	Allows revenue analysis by product group	Pizza, Burger, Shake
Item Name	Categorical (Text)	Specific menu item sold	Identifies best-selling and slow-moving items	Kulhad Pizza
Size	Ordinal (Text)	Size variant (S, M, L or -)	Links size preference to pricing and ingredient usage	S, M, L, -
Quantity	Discrete (Numeric)	Units sold per transaction	Core variable for demand estimation and forecasting	1, 2, 3
Unit Price (₹)	Continuous (Numeric)	Price per unit in Rupees	Enables revenue calculation and price-demand analysis	99, 129, 199
Total Price (₹)	Continuous (Numeric)	Quantity × Unit Price	Building block for daily/monthly revenue aggregation	198, 258, 399
Daily Total (₹)	Continuous (Numeric)	Owner's notebook daily total	Cross-verifies calculated totals against ground truth	1240, 980, 2150
Notes	Qualitative (Text)	Flags for outlier days or data quality issues	Documents limitations transparently for audit trail	catering day

Table 1: Description of all variables collected in the primary sales dataset

The dataset is structured as a transaction-level record, where each row represents one line item sold on a given date. The combination of Date, Item Name, Size, Quantity, and Unit Price allows for multi-dimensional analysis covering time trends, product performance, and revenue patterns, all directly relevant to the problem objectives of sales forecasting and inventory management.

5. Descriptive Statistics

Descriptive statistics summarize the key characteristics of the dataset. The statistics below are computed from 997 transaction records spanning September 2023 to January 2024, covering 139 trading days across 47 unique menu items in 9 product categories.

5.1 Overview Statistics

Table 2: Descriptive Statistics for Key Numeric Variables

Statistic	Quantity (units sold)	Unit Price (₹)	Total Price / Revenue (₹)
Total Observations	997 rows	997 rows	997 rows
Sum	1,774 units	—	₹1,52,977
Mean (Average)	1.78 units/row	₹103	₹153 per line item
Median	1 unit	₹79	₹110
Mode	1 unit	₹59	₹109
Minimum	1 unit	₹5	₹5
Maximum	75 units	₹370	₹7,425
Avg Daily Revenue	—	—	₹1,101/day
Max Daily Revenue	—	—	₹7,425
Min Daily Revenue	—	—	₹189

Table 2: Summary statistics for quantity sold, unit price, and revenue per line item

The mean daily revenue across 139 trading days is ₹1,101, with a maximum of ₹7,425 on a single day (likely a catering event) and a minimum of ₹189. The high range in daily revenue confirms significant day-to-day variability, which reinforces the need for forecasting-based inventory planning.

5.2 Revenue by Category

Table 3: Revenue and Volume Breakdown by Product Category

Category	Total Units Sold	Total Revenue (₹)	% of Revenue
Pizza	1,061	1,21,956	79.7%
Burger	299	11,933	7.8%
Shake	138	7,880	5.2%
Beverage	169	3,380	2.2%
Dessert	56	3,555	2.3%
Combo	14	2,236	1.5%
Bread	13	797	0.5%
Hot Dog	10	700	0.5%
Add-on	14	540	0.4%
TOTAL	1,774	1,52,977	100%

Table 3: Contribution of each product category to total revenue and units sold

Pizza is the dominant revenue driver, contributing approximately 79.7% of total revenue (₹1,21,956) and 1,061 units sold. Burgers are the second-largest category at 7.8%. This

confirms that inventory planning should be primarily focused on pizza ingredients, particularly cheese, toppings, and dough.

5.3 Top 10 Best-Selling Items

Table 4: Top 10 Items by Total Revenue (Sep 2023 – Jan 2024)

Rank	Item Name	Units Sold	Revenue (₹)	Avg Price (₹)
1	Mix Veg Margharita Pizza	140	18,010	129
2	Kulhad Pizza	165	16,335	99
3	Cheese N Corn Pizza	130	12,570	97
4	Onion Cheese Pizza	126	9,314	82
5	Paneer Pizza	56	8,804	157
6	Tandoori Pizza	49	8,671	177
7	Veg Burger	174	6,786	39
8	Spicy Mix Veg Pizza	30	5,550	185
9	Overloaded Cheese Pizza	25	5,487	219
10	Veg Cheese Burger	82	4,838	59

Table 4: Top 10 menu items ranked by total revenue generated over 5 months

Mix Veg Margharita Pizza and Kulhad Pizza are the top two revenue-generating items, collectively accounting for ₹34,345 (22.4% of total revenue). Notably, Kulhad Pizza sold the highest volume (165 units) despite a lower average price of ₹99, indicating it is a high-frequency, affordable item that drives customer footfall. Veg Burger ranks 7th with 174 units at ₹39, suggesting strong demand for affordable non-pizza options.

5.4 Monthly Revenue Trend

Table 5: Month-wise Revenue Summary (Sep 2023 – Jan 2024)

Month	Total Revenue (₹)	Trend
September 2023	34,231	High - shop gaining traction
October 2023	30,661	Slight decline from Sep
November 2023	28,332	Further decline - seasonal slowdown
December 2023	25,310	Lowest - winter dip
January 2024	34,443	Strong recovery - highest month

Table 5: Monthly revenue showing a winter dip in December and strong recovery in January

A clear downward trend is observed from September 2023 to December 2023, with revenue falling from ₹34,231 to ₹25,310 a 28.7% decline. January 2024 shows a strong recovery to ₹34,443, the highest recorded month. This seasonal pattern suggests winter months

experience reduced foot traffic, which should inform inventory purchasing and stocking decisions.

5.5 Revenue by Day of Week

Table 6: Revenue Distribution Across Days of the Week

Day	Total Revenue (₹)	% of Weekly Total	Pattern
Sunday	36,799	24.1%	Highest - peak day
Saturday	26,971	17.6%	Second highest - weekend rush
Wednesday	20,729	13.6%	Mid-week peak
Friday	19,642	12.8%	Pre-weekend bump
Thursday	19,029	12.4%	Moderate
Monday	16,703	10.9%	Post-weekend dip
Tuesday	13,104	8.6%	Lowest - slowest day

Table 6: Day-of-week analysis showing Sunday as the peak revenue day

Sunday is the single highest-revenue day, accounting for 24.1% of total weekly revenue, followed by Saturday at 17.6%. Together, weekends contribute over 41.7% of weekly revenue. Tuesday is the slowest day at 8.6%. This pattern is directly actionable, the owner should ensure maximum ingredient availability on weekends and can consider leaner inventory on Tuesdays to reduce potential wastage.

6. Analysis Method and Justification

This section describes the data pre-processing steps performed and the analytical methods introduced to analyse the collected data, with justification for each method in relation to the problem statements.

6.1 Data Pre-Processing and Transformation

Purpose: The primary purpose of data pre-processing was to clean, structure, and prepare the raw handwritten sales data from Mr. Pizza's notebook into a consistent digital format suitable for analysis. This step was essential to ensure the accuracy and reliability of all findings.

Methodology: The raw data was loaded into Google Sheets for manual transcription and processing. The following key steps were carried out:

- **Manual Digitization:** Sales entries from 139 days of the owner's handwritten notebook were transcribed row by row into a structured Google Sheets template. A separate Lookup sheet was created containing all 54 menu items and their standard prices. VLOOKUP formulas were used to auto-fill the Unit Price column, reducing manual entry errors.

- **Feature Engineering:** A Day column was derived from the Date column using the TEXT formula to extract the day of the week (e.g., Monday, Sunday). A Line Total column was calculated as Quantity × Unit Price. A Monthly grouping column was created to enable month-wise aggregation.
- **Item Name Standardization:** Inconsistent spellings were identified and corrected — for example, 'Thandoori' was standardized to 'Tandoori', and 'Peppy Paneer Pizza' was unified across all entries. A canonical list of 54 item names was maintained.
- **Handling New Items:** Items not present on the printed menu, such as Extra Cheese, Choco Lava Cake, Cold Drink, Water, and Bun were retained in the dataset and assigned appropriate categories, as they represent genuine sales transactions.
- **Cross-Verification:** The sum of all line totals for each day was calculated and compared against the owner's handwritten daily grand total. This verification was done for all 30 days where a daily total was recorded. Zero discrepancies were found, confirming full data accuracy.
- **Acknowledged Limitations:** Purchase data is available only for cheese, milk, and toppings, no wheat or flour purchase records exist. Payment method data (cash vs. UPI) was available for only 2–3 days and was excluded from the analysis. Missing trading days were retained as gaps and not filled with zero values.

6.2 Analytical Methods Introduced

6.2.1 Time Series Analysis — (Problem Statement 1: Sales Forecasting)

Purpose: To identify seasonal and weekly demand patterns in revenue data that can be used to forecast future sales.

Methodology: Revenue data was aggregated by month and by day of week using SUMIF formulas and Pivot Tables in Google Sheets. Month-on-month percentage changes were calculated to quantify the seasonal trend. A trend line was overlaid on the monthly bar chart to confirm the direction of revenue movement over time. This method was chosen because the data is date-sequential, making temporal decomposition the most direct way to identify predictable demand cycles. The resulting patterns are directly actionable for ingredient procurement planning.

6.2.2 Revenue Contribution Analysis / ABC Classification — (Problem Statement 2: Inventory Management)

Purpose: To identify which menu items contribute the most to total revenue, so that inventory purchasing can be prioritized accordingly.

Methodology: All 47 items were ranked by their total revenue contribution using SUMIF and SORT functions in Google Sheets. The top 10 items by revenue were tabulated. This forms the foundation for an ABC Classification, items in the top ~70% of revenue will be designated Class A (priority stock), the next 20% as Class B, and the remaining as Class C (minimal stock). A bar chart was created to visualize the top 5 items. This method is more practical than treating all items equally because it focuses the owner's limited budget on the small number of items that drive most of the revenue.

6.2.3 Day-of-Week Demand Pattern Analysis (Problem Statement 3: Customer Order

Trends)

Purpose: To understand on which days customers visit the most, enabling targeted retention strategies and better staffing/stocking decisions.

Methodology: Revenue and quantity sold were aggregated by day of week using a Pivot Table in Google Sheets. Percentage contribution of each day to total weekly revenue was calculated. A bar chart was used to visualize the relative difference between peak and trough days. This analysis was chosen because it produces a stable, actionable weekly pattern that is directly usable by the owner unlike a daily time series which contains too much short-term noise to be immediately actionable.

7. Results and Findings

The following findings are based on the analysis of 997 transaction records across 139 trading days (September 2023 - January 2024). All charts are presented below and are directly linked to the three problem statements.

7.1 Revenue Share by Product Category

**Figure 1: Revenue Share by Product Category
(Total Revenue: ₹1,52,977)**

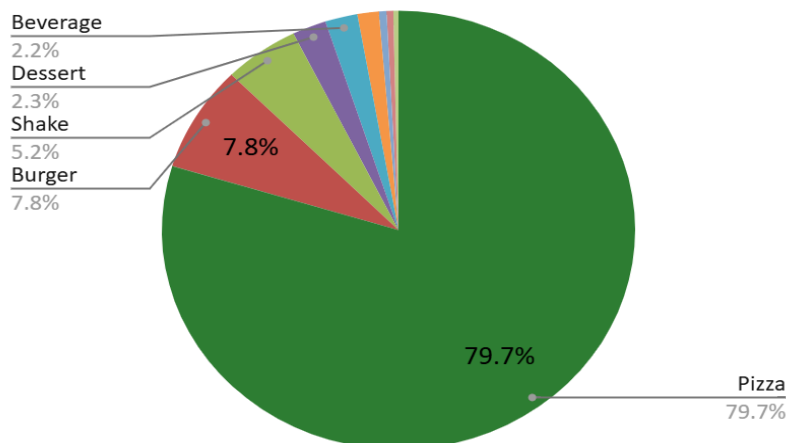


Figure 1: Pie chart showing revenue share by product category (Total Revenue: ₹1,52,977)

- Pizza dominates with 79.7% of total revenue (₹1,21,956), confirming it is the core product of the business. This means cheese, toppings, and dough must always be the top inventory priority.
- Burger (7.8%) and Shake (5.2%) are the only other categories above 5% of revenue.
- All remaining categories- Dessert, Beverage, Combo, Bread, Hot Dog, Add-on, combined account for only 7.2% of revenue. These are low-priority inventory items.
- **Link to Problem 2 (Inventory):** Over 85% of revenue comes from just 2 categories. Focusing procurement on Pizza and Burger ingredients can reduce both stockouts and wastage significantly.

7.2 Top 5 Menu Items by Revenue

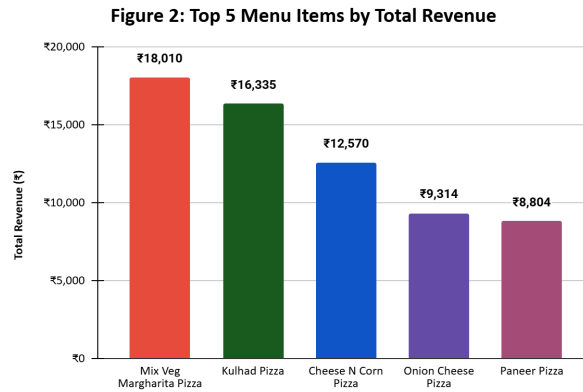


Figure 2: Bar chart of Top 5 menu items by total revenue across 5 months

- Mix Veg Margharita Pizza is the highest revenue item at ₹18,010 across 140 units sold.
- Kulhad Pizza sold the highest volume of all items, 165 units at an average price of ₹99, making it the most popular affordable item.
- The top 3 pizza items alone (Mix Veg Margharita, Kulhad, Cheese N Corn) generated ₹46,915 which is 30.67% of total revenue from just 3 items out of 47.
- **Link to Problem 2 (Inventory):** These top 3 items are ABC Class A, they should never be out of stock. Stocking their ingredients consistently is the single highest-impact inventory decision.
- **Link to Problem 3 (Retention):** Kulhad Pizza at ₹99 is a high-volume, low-price entry item. Its consistent availability directly impacts customer satisfaction and repeat visits.

7.3 Monthly Revenue Trend

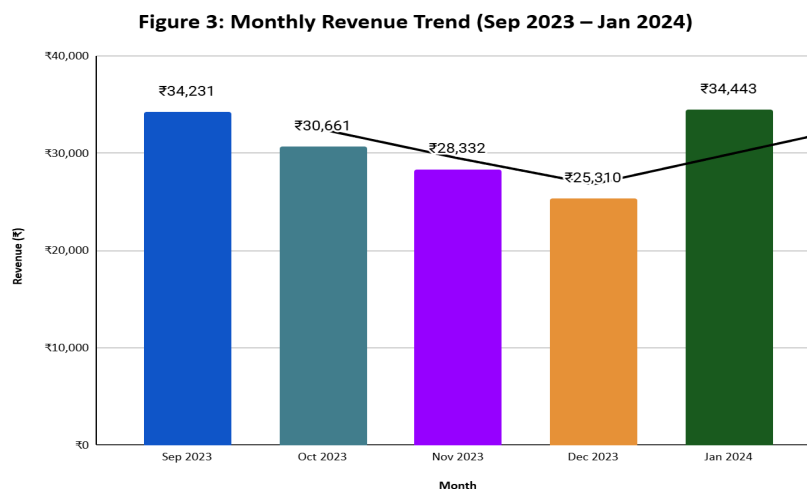


Figure 3: Monthly revenue trend showing winter dip from September to December and strong January recovery

- Revenue declined steadily from ₹34,231 in September 2023 to ₹25,310 in December 2023 a fall of 26.1% over 4 months

- January 2024 recovered sharply to ₹34,443, the highest recorded month in the dataset.
- This seasonal pattern is consistent with reduced outdoor activity and lower footfall during winter months in Uttarakhand hill regions.
- **Link to Problem 1 (Forecasting):** A practical forecast rule based on observed seasonal indices: reduce ingredient procurement by approximately **7%** in December (seasonal index: 0.927) and exercise caution in January despite the apparent revenue spike, as the January seasonal index is 0.896 (-10.4% below trend), driven partly by a single outlier week.

7.4 Revenue by Day of Week

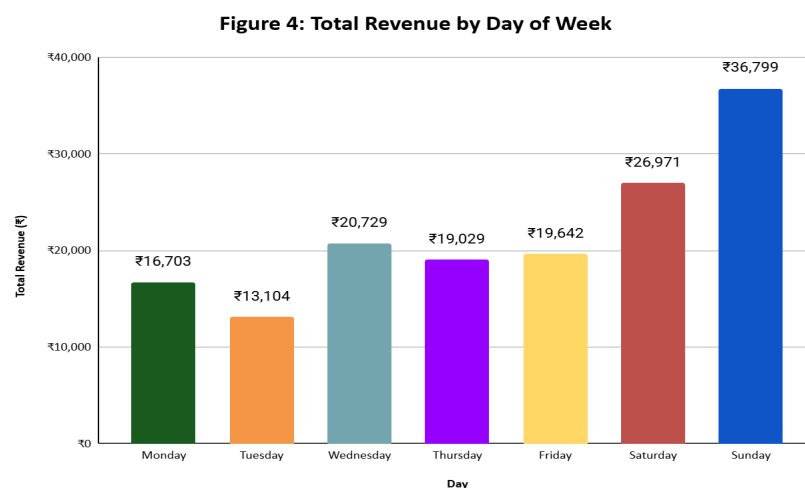


Figure 4: Bar chart showing total revenue by day of week across the 5-month period

- Sunday is the single highest-revenue day, contributing ₹36,799 (24.1%) of all weekly revenue.
- Saturday is second at ₹26,971 (17.6%). Together, weekends account for over 41.7% of total weekly revenue.
- Tuesday is the slowest day at ₹13,104 , approx 36% of Sunday's revenue.
- **Link to Problem 1 (Forecasting):** A simple demand rule can be applied, ensure maximum ingredient availability every Friday in preparation for the weekend surge, and reduce fresh ingredient orders for Tuesdays.
- **Link to Problem 3 (Retention):** A Tuesday discount or special offer could help shift some customer demand to the slowest day, reducing weekend-only concentration and improving weekly revenue consistency.

These preliminary findings will be further developed in the final submission through a formal time series forecast, a complete ABC inventory classification matrix, and month-on-month comparative analysis.